SkillForge

Data Engineering Department · Task Plan

This is the Data Engineering team's working plan for SkillForge V1, from the first commit to the production launch, and the work that follows it. It lists every Data Engineering task, what must be ready before each one, how to prove it is done, and how the work reaches production. The **SkillForge Project Specification v2.1** remains the source of truth for architecture and scope.

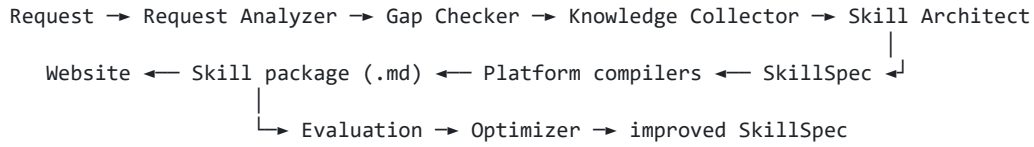
Department	Data Engineering (DE)
Tasks	16 tasks: DE-01 to DE-16
V1 timeline	26 October 2026 – 12 February 2027 (16 weeks, proposed)
Companion document	SkillForge Project Specification v2.1
Date	22 October 2026

Your work at a glance

Phase	Tasks	IDs
P0 · Foundations (Weeks 1–2)	3	DE-01–03
P1 · Walking skeleton (Weeks 3–6)	3	DE-04–06
P2 · MVP features (Weeks 7–10)	4	DE-07–10
P3 · Hardening & beta (Weeks 11–13)	3	DE-11–13
P4 · Production launch (Weeks 14–16)	2	DE-14–15
P5 · After launch (V2 / V3)	1	DE-16

1. About SkillForge

SkillForge is a website and AI pipeline that creates **skills** for AI agents. A user names any skill they need, technical or non-technical (for example AI Engineer or Marketing Strategist). If no good skill exists, SkillForge researches the domain, designs a validated **SkillSpec** and generates an installable Markdown package (`SKILL.md` plus reference files). The package drops into Claude Code (`.claude/skills/`), Codex (`.agents/skills/`) and later other agents, which then act as that skill.



Where Data Engineering fits: The crawling and processing systems behind the Knowledge Collector, the knowledge store and retrieval index, provenance, evaluation datasets, analytics and data privacy.

2. The Data Engineering Department

Builds the systems that collect web knowledge and public skills responsibly, process them into clean, searchable passages with full provenance, and keep datasets, analytics and user data in order.

Item	Details
Owns	Crawling service, processing pipeline, knowledge schemas, pgvector indexes, object storage layout, provenance, public skill corpus, evaluation datasets, analytics, data privacy
Technology	Python, httpx, Playwright, trafilatura, MinHash, PostgreSQL + pgvector, object storage, job scheduler, data-quality checks
Works most with	AI (extraction rules, retrieval), Backend (database, storage), DevOps (infrastructure, scraping network)

> **Boundary between AI and Data Engineering:** AI decides what to collect and how to interpret it (source selection, extraction rules, chunking, embeddings, retrieval quality). Data Engineering builds and runs the systems that collect, process and store it at scale (crawler, pipelines, storage, provenance, compliance). Web scraping quality is an AI responsibility. Scraping infrastructure is a Data Engineering responsibility.

All departments

Department	Mission
Frontend (FE)	Make skill creation simple for technical and non-technical users
Backend (BE)	Run the product reliably and turn SkillSpecs into skill packages
AI	Turn a request plus knowledge into a correct, grounded, useful skill
Data Engineering (DE)	Collect, store and serve trustworthy data at scale
DevOps (OPS)	Ship safely and keep production healthy

3. Timeline

Week 1 starts on Monday 26 October 2026. A phase is finished only when its exit criteria are met by all departments. Dates will move if the team plans around the December–January holidays.

Phase	Weeks and dates	Exit criteria (all departments)	Your tasks
P0 · Foundations	Weeks 1–2 26 Oct – 6 Nov 2026	SkillSpec frozen (5 Nov), OpenAPI v1 merged, CI green, every developer runs the stack locally	DE-01–03
P1 · Walking skeleton	Weeks 3–6 9 Nov – 4 Dec 2026	On staging: request → job → AI-generated SkillSpec (no web knowledge yet) → Claude package → download	DE-04–06
P2 · MVP features	Weeks 7–10 7 Dec 2026 – 1 Jan 2027	Web knowledge with citations, Codex compiler, library, preview, editing, ratings working on staging	DE-07–10
P3 · Hardening & beta	Weeks 11–13 4 – 22 Jan 2027	Tests, security review, load test and AI quality gates pass. Beta users create 50+ skills	DE-11–13
P4 · Production launch	Weeks 14–16 25 Jan – 12 Feb 2027	Go-live checklist (§8) signed by all five departments	DE-14–15
P5 · After launch	From Week 17	Per V2/V3 goals above	DE-16

4. How to Work

- **Task IDs.** Every task has an ID: `FE-`, `BE-`, `AI-`, `DE-` or `OPS-` plus a number. Create one GitHub issue per task with the ID in its title, and name branches after it (e.g. `fe-05-create-wizard`).
- **Phases.** Tasks are grouped by the phases in §3. A department may start a task early if its dependencies are ready, but a phase is finished only when its exit criteria are met by all departments.
- **"Needs" column.** The tasks (often from other departments) that must be ready first. A dash means none. Blocked work is raised at the weekly cross-department sync, not left waiting.
- **"Done when" column.** The check that proves the task is finished. A task is not done because the code is written. It is done when this check passes.
- **Weekly rhythm.** Monday planning per department, Thursday cross-department sync (blockers, contract changes), Friday demo on staging from P1 onwards.

Definition of Done (every task)

1. Code is in a pull request linked to the task issue, reviewed and approved by the owning department (and by consuming departments if a contract changed).
2. Tests are added or updated, and CI is green: lint, type checks, tests, security scans.
3. No secrets, keys or personal data in the code or logs.
4. New endpoints, jobs and pipelines emit logs, metrics and traces.
5. Documentation is updated (README, API docs or docs/ page).
6. The change is deployed to staging and checked there. From P4 onwards it also follows the release process in §8.

5. Contracts You Produce or Use

A contract changes only through a pull request approved by every department that consumes it.

Contract	Your role	Producer	Consumers	Task	Due
Database schemas and migrations	Producer and consumer	BE (app data), DE (knowledge data)	All	BE-02, DE-01	Week 2
Environment variables, Docker images, local setup	Consumer	OPS	All	OPS-03	Week 2
Knowledge retrieval: query + domain → passages with sources	Producer	DE + AI	AI	DE-05, AI-11	Week 8
Analytics event schema	Producer	DE	FE, BE	DE-12	Week 10

6. Dependencies on Other Departments

Track these at the Thursday cross-department sync. If a task you need is late, raise it there.

6.1 What you need from others

Their task	Department	What it delivers	Phase	Needed by
AI-04	AI	Seed benchmark: 10 seed skills (technical and non-technical) with test prompts and rubrics	P0	DE-10
BE-04	Backend	Authentication: email and OAuth (GitHub / Google), user and admin roles	P0	DE-13
OPS-03	DevOps	Local development: Dockerfiles and docker-compose (api, worker, web, PostgreSQL + pgvector, Redis)	P0	DE-02, DE-03
OPS-09	DevOps	Scraping network: separate worker pool, egress control, outbound rate limits	P1	DE-04
OPS-10	DevOps	Observability: OpenTelemetry collector, logs, metrics, traces, dashboards, alerts (errors, queue depth, LLM spend)	P2	DE-11
OPS-12	DevOps	Backups and disaster-recovery runbook	P2	DE-14
BE-18	Backend	Observability: structured logs, traces through the whole job, metrics (job time, failures, cost per job)	P3	DE-12
AI-23	AI	Optimizer: analyze failures, mutate SkillSpec fields, keep the best version (V3)	P5	DE-16

6.2 What others need from you

Your task	What it delivers	Phase	Needed by
DE-01	Data architecture: knowledge schemas (sources, documents, chunks, embeddings) and retention rules	P0	BE-02
DE-02	PostgreSQL + pgvector setup and index strategy (HNSW)	P0	BE-11, AI-11

Your task	What it delivers	Phase	Needed by
DE-03	Object storage layout for raw pages, clean text and packages; lifecycle rules	P0	BE-10
DE-04	Crawling service: httpx and Playwright fetchers, robots.txt and terms checks, per-domain rate limits, identified user agent, retries	P1	AI-09, AI-10
DE-08	Public skill corpus: collect public SKILL.md files, detect licenses, store metadata	P2	AI-08
DE-12	Analytics: event schema and dashboards (requests, success rate, cost, ratings)	P3	FE-21, AI-24

7. Task Plan

Tasks in phase order. Create one GitHub issue per task with its ID in the title.

P0 · Foundations (Weeks 1–2)

ID	Task	Done when	Needs
DE-01	Data architecture: knowledge schemas (sources, documents, chunks, embeddings) and retention rules	Reviewed by BE and AI	—
DE-02	PostgreSQL + pgvector setup and index strategy (HNSW)	Similarity query under 100 ms on a 1M-chunk test set (local first, staging in P1)	OPS-03
DE-03	Object storage layout for raw pages, clean text and packages; lifecycle rules	Layout documented; applied to staging storage in P1	OPS-03

P1 · Walking skeleton (Weeks 3–6)

ID	Task	Done when	Needs
DE-04	Crawling service: httpx and Playwright fetchers, robots.txt and terms checks, per-domain rate limits, identified user agent, retries	Disallowed URLs are never fetched (tested)	OPS-09
DE-05	Processing pipeline: raw → clean → chunk → embed; idempotent, versioned jobs. Chunking and embedding settings are configurable so AI can tune them in AI-10 and AI-11	Re-running a job gives the same output	DE-04
DE-06	Provenance: URL, retrieval date, license and content hash stored with every chunk	Every chunk traceable to its source	DE-05

P2 · MVP features (Weeks 7–10)

ID	Task	Done when	Needs
DE-07	Deduplication (MinHash), language detection, quality filters	Duplicate rate reported per crawl	DE-05
DE-08	Public skill corpus: collect public SKILL.md files, detect licenses, store metadata	Only permissively licensed files used as examples	DE-04
DE-09	Domain knowledge cache: reuse documents across skills, freshness rules, re-crawl schedule	A second request in the same domain reuses the cache	DE-05
DE-10	Evaluation datasets: versioned benchmark sets and human labels	Every evaluation run records its dataset version	AI-04

P3 · Hardening & beta (Weeks 11–13)

ID	Task	Done when	Needs
DE-11	Data-quality checks and pipeline monitoring (failures, volumes, freshness)	Alert on failed or empty runs	OPS-10
DE-12	Analytics: event schema and dashboards (requests, success rate, cost, ratings)	Dashboard used in the weekly review	BE-18
DE-13	Privacy: personal data in requests, user data deletion, retention enforcement	Deletion request removes user data within the agreed time	BE-04

P4 · Production launch (Weeks 14–16)

ID	Task	Done when	Needs
DE-14	Production pipelines scheduled; database and vector backup and restore tested with DevOps	Restore drill passed	OPS-12
DE-15	Capacity plan: storage, embeddings, crawl volume for the first 6 months	Reviewed with DevOps	—

P5 · After launch (V2 / V3)

ID	Task	Done when	Needs
DE-16	Data for the optimizer and experiment tracking (MLflow)	Every optimizer run reproducible from stored data	AI-23

8. From Code to Production

Every change follows the same path. DevOps owns the pipeline. Your department owns the change.

Environments

Environment	Purpose	Deployed when	Data
Local	Daily development (docker-compose)	Developer runs it	Sample data only
Preview	Review a frontend PR in the browser	Every FE pull request	Staging API
Staging	Integration, demos, QA, beta	Every merge to main	Test and beta data
Production	Real users	Approved release tag only	Real data, backed up

Release steps

#	Step	Who	Gate
1	Pick a task issue (e.g. BE-09) and create a branch named after it	Developer	Task is in the current phase
2	Code, with tests and docs	Developer	Shared Definition of Done (§4)
3	Open a pull request; CI runs lint, types, tests, security and secret scans	Developer, CI	CI green
4	Review by the owning department (and by consumers if a contract changed)	Reviewers	Approval via CODEOWNERS
5	Merge to main → automatic deploy to staging with migrations	CI/CD	Deploy succeeds; smoke tests pass
6	Verify on staging: E2E tests, manual QA, AI quality gates for AI changes	Owning department	All checks pass
7	Create release candidate: version tag `vX.Y.Z` and changelog	DevOps	Release notes reviewed
8	Production approval	DevOps + one reviewer per department	No open critical bugs or security findings
9	Deploy to production (migrations first, then rolling or blue-green)	DevOps	Health checks pass
10	Watch dashboards for one hour; roll back with one command if alerts fire	DevOps + on-call	Error rate and latency normal

Your go-live checks (signed in OPS-19)

Check	Owner	Evidence
Crawler compliance tests pass; every chunk has provenance	DE	DE-04, DE-06 tests
Backups restored successfully; data deletion works	DE + OPS	DE-13, DE-14, OPS-12